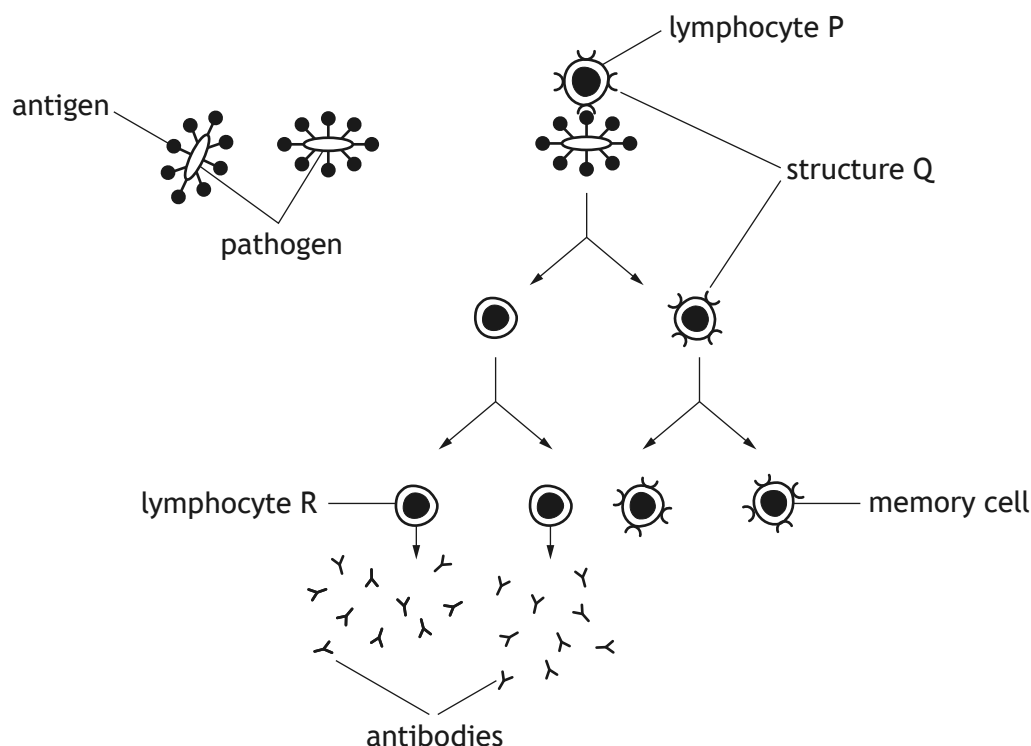


13. The diagram represents lymphocyte responses to infection by a pathogen.



- (a) (i) Structure Q is found on the surface of lymphocytes to allow them to bind with antigens.

Name structure Q.

1

- (ii) Give evidence from the diagram that shows lymphocyte R is a B lymphocyte.

1

- (b) Antibodies attach to and inactivate a pathogen, so the pathogen cannot spread through the body.

Describe the role of phagocytes following the inactivation of the pathogen.

1

[Turn over

13. (continued)

- (c) An individual contracted chicken pox during childhood. A few years later they were exposed to the same chicken pox antigen.

Describe the role of memory cells in protecting this individual from developing symptoms of chicken pox during this second exposure.

2

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- (d) T lymphocytes are another type of lymphocyte that protect the body from infection.

Explain how T lymphocytes destroy infected cells.

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